Lecture 7 - Java Graphical User Interface (GUI): JavaFX - Part III

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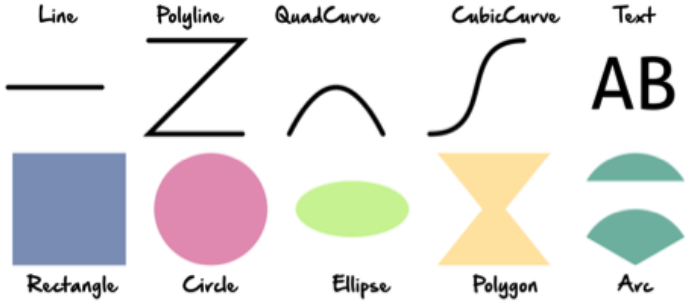
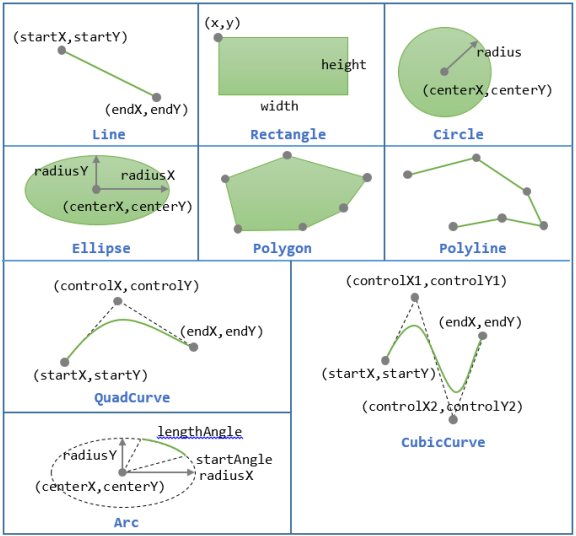
CSC-1004: Computational Laboratory Using Java
Course Page: [Click]

JavaFX 2D Shapes

JavaFX provides the flexibility to create our 2D shapes on the screen. Various classes can be used to implement 2D shapes in our application. All these classes reside in the javafx.scene.shape package.



JavaFX 2D Shapes



JavaFX 2D Shapes: Line

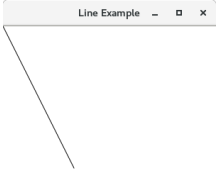
JavaFX library allows the developers to create a Line.

```
public class LineDrawingExamples extends Application{

  @Override
  public void start(Stage primaryStage) throws Exception {
    // TODO Auto-generated method stub
    Line line = new Line(); //instantiating Line class
    line.setStartX(0); //setting starting X point of Line
    line.setStartY(0); //setting starting Y point of Line
    line.setEndX(100); //setting ending X point of Line
    line.setEndY(200); //setting ending Y point of Line
    Group root = new Group(); //Creating a Group
    root.getChildren().add(line); //adding the class object //to the group
    Scene scene = new Scene(root,300,300);
    primaryStage.setScene(scene);
    primaryStage.setTitle("Line Example");
    primaryStage.show();

  }
  public static void main(String[] args) {
    launch(args);
  }

}
```



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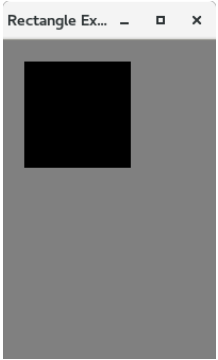
JavaFX 2D Shapes: Rectangle

JavaFX library allows the developers to create a rectangle.

```
public class Shape_Example extends Application{

  @Override
  public void start(Stage primaryStage) throws Exception {
    // TODO Auto-generated method stub
  primaryStage.setTitle("Rectangle Example");
  Group group = new Group(); //creating Group
  Rectangle rect=new Rectangle(); //instantiating Rectangle
  rect.setX(20); //setting the X coordinate of upper left //corner of rectangle
  rect.setY(20); //setting the Y coordinate of upper left //corner of rectangle
  rect.setWidth(100); //setting the width of rectangle
  rect.setHeight(100); // setting the height of rectangle
  group.getChildren().addAll(rect); //adding rectangle to the //group
  Scene scene = new Scene(group,200,300,Color.GRAY);
  primaryStage.setScene(scene);
  primaryStage.show();
}
public static void main(String[] args) {
  launch(args);
}

}
```



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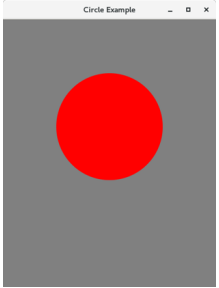
JavaFX 2D Shapes: Circle

JavaFX library allows the developers to create a Circle.

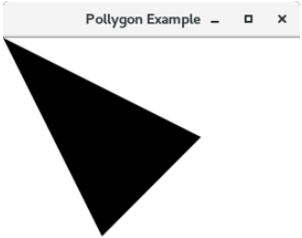
```
public class Shape_Example extends Application{

    @Override
    public void start(Stage primaryStage) throws Exception {
        // TODO Auto-generated method stub
    primaryStage.setTitle("Circle Example");
    Group group = new Group();
    Circle circle = new Circle();
    circle.setCenterX(200);
    circle.setCenterY(200);
    circle.setRadius(100);
    circle.setFill(Color.RED);
    group.getChildren().addAll(circle);
    Scene scene = new Scene(group,400,500,Color.GRAY);
    primaryStage.setScene(scene);
    primaryStage.show();
}
public static void main(String[] args) {
    launch(args);
}

}
```



JavaFX 2D Shapes: Polygons

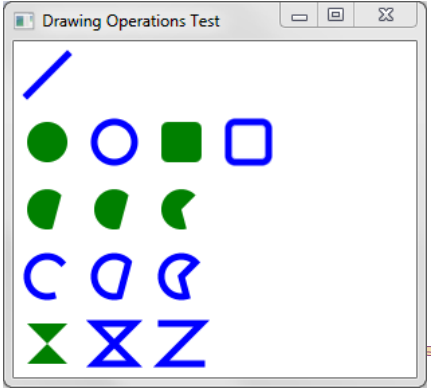
In JavaFX, Polygon can be created by instantiating javafx.scene.shape.Polygon class.

```java
public class Shape_Example extends Application {

 @Override
 public void start(Stage primarystage) {
  Group root = new Group();
  primarystage.setTitle("Pollygon Example");

  Polygon polygon = new Polygon();
  polygon.getPoints().addAll(new Double[]{
     0.0, 0.0,
     100.0, 200.0,
     200.0, 100.0 });

  root.getChildren().add(polygon);
  Scene scene = new Scene(root,300,400);
  primarystage.setScene(scene);
  primarystage.show();
 }

 public static void main(String[] args) {
  launch(args);
 }
}
```



JavaFX Canvas

Canvas class creates an image that can be drawn on using a set of graphics commands provided by a GraphicsContext. Canvas has a specified height and width and all the drawing operations are clipped to the bounds of the canvas.

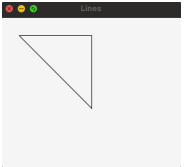


JavaFX Canvas

In the first example, we draw simple lines.

```java
public class CanvasLine extends Application {

    @Override
    public void start(Stage stage) {

        initUI(stage);
    }

    1 usage
    private void initUI(Stage stage) {

        var root = new Pane();

        var canvas = new Canvas( v: 300,  v1: 300);
        var gc = canvas.getGraphicsContext2D();
        drawLines(gc);

        root.getChildren().add(canvas);

        var scene = new Scene(root,  v: 300,  v1: 250, Color.WHITESMOKE);

        stage.setTitle("Lines");
        stage.setScene(scene);
        stage.show();
    }
```

```java
    1 usage
    private void drawLines(GraphicsContext gc) {

        gc.beginPath();
        gc.moveTo( v: 30.5,  v1: 30.5);
        gc.lineTo( v: 150.5,  v1: 30.5);
        gc.lineTo( v: 150.5,  v1: 150.5);
        gc.lineTo( v: 30.5,  v1: 30.5);
        gc.stroke();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```



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JavaFX Canvas

- var canvas = new Canvas(300, 300); A Canvas is constructed with a width and height that specifies the size of the image into which the canvas drawing commands are rendered.
- var gc = canvas.getGraphicsContext2D(); The getGraphicsContext2D returns a GraphicsContext associated with the canvas.
- gc.beginPath(); A line primitive is represented as a path element. The beginPath method starts a new path.

JavaFX Canvas

- gc.moveTo(30.5, 30.5); The moveTo method moves the starting point of the current path to the specified coordinate.
- gc.lineTo(150.5, 30.5); The lineTo methods add line segments to the current path.
- gc.stroke(); The stroke method strokes the path with the current stroke paint.

JavaFX Canvas

A stroke is used to draw outlines of shapes. A fill is used to paint interiors of shapes.

```java
public class CanvasStrokeFill extends Application {

    @Override
    public void start(Stage stage) {
        var root = new Pane();
        var canvas = new Canvas( v: 300,  v1: 300);
        var gc = canvas.getGraphicsContext2D();
        doDrawing(gc);
        root.getChildren().add(canvas);
        var scene = new Scene(root,  v: 300,  v1: 250, Color.WHITESMOKE);
        stage.setTitle("Stroke and fill");
        stage.setScene(scene);
        stage.show();
    }

    1 usage
    private void doDrawing(GraphicsContext gc) {
        gc.setStroke(Color.FORESTGREEN.brighter());
        gc.setLineWidth(5);
        gc.strokeOval( v: 30,  v1: 30,  v2: 80,  v3: 80);
        gc.setFill(Color.FORESTGREEN);
        gc.fillOval( v: 130,  v1: 30,  v2: 80,  v3: 80);
    }

    public static void main(String[] args) { launch(args); }
}
```

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JavaFX Canvas

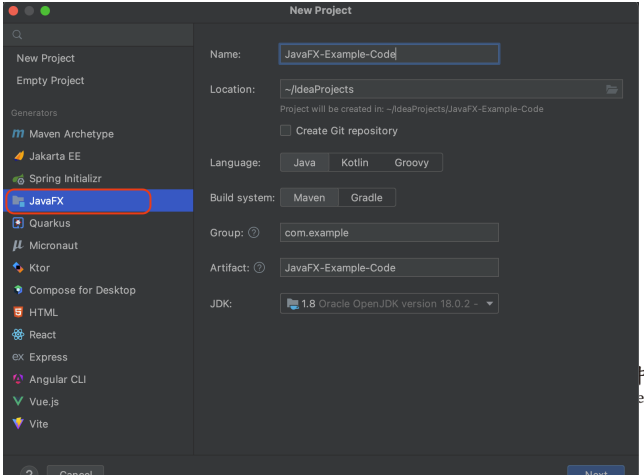
- gc.setStroke(Color.FORESTGREEN.brighter()); The setStroke method sets the current stroke paint attribute. The default colour is black. The attribute is used by the stroke methods of the GraphicsContext.
- gc.setLineWidth(5); The setLineWidth sets the current line width.
- gc.strokeOval(130, 30, 80, 80); The strokeOval method strokes an oval using the current stroke paint.

JavaFX Canvas

- gc.setFill(Color.FORESTGREEN); The setFill method sets the current fill paint attribute. The default colour is black. The attribute is used by the fill methods of the GraphicsContext.
- gc.fillOval(30, 30, 80, 80); The fillOval fills an oval using the current fill paint.

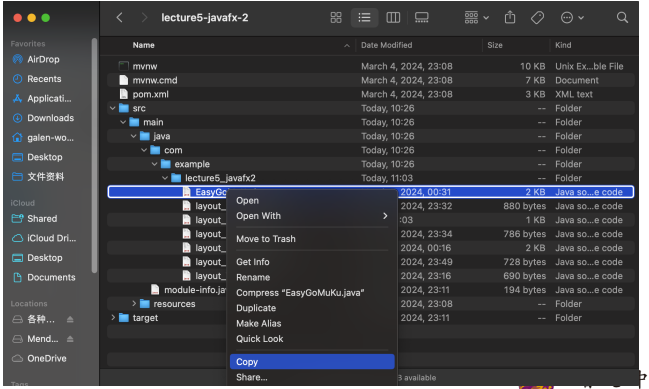
Import JavaFX Example Code

- Step 1: Create a new JavaFX Project on your PC/Laptop. Try to use *File->New->Project* under the Intellij idea.



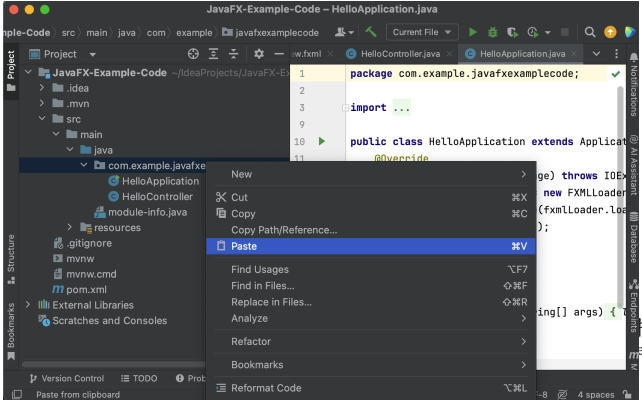
Import JavaFX Example Code

- Step 2: Copy the target .Java File from the folder of example code.



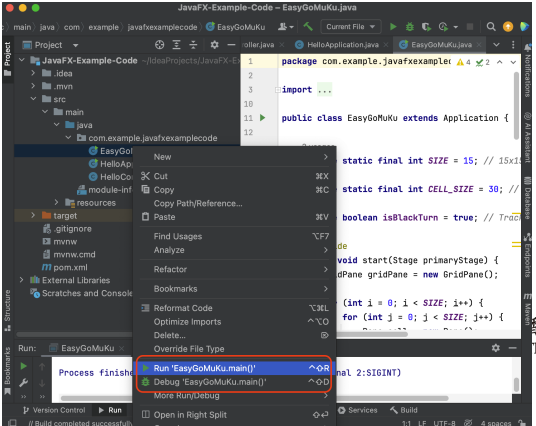
Import JavaFX Example Code

- Step 3: Paste the .Java File to the folder of the right place (e.g., *YourProjectName/src/java/com/example/YourFolderName*) in the created project. The pasting must be done with the Intellij idea.



Import JavaFX Example Code

- Step 4: Run/Debug the code by right-clicking the Java file (config your JDK if necessary, and please check whether the JavaFX package has been downloaded to the "External Libraries" in your project.).



Question and Answering (Q&A)